

Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

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Problem formulation

The mass spectrometry analysis includes two approach: **targeted** analysis and **untargeted** analysis.

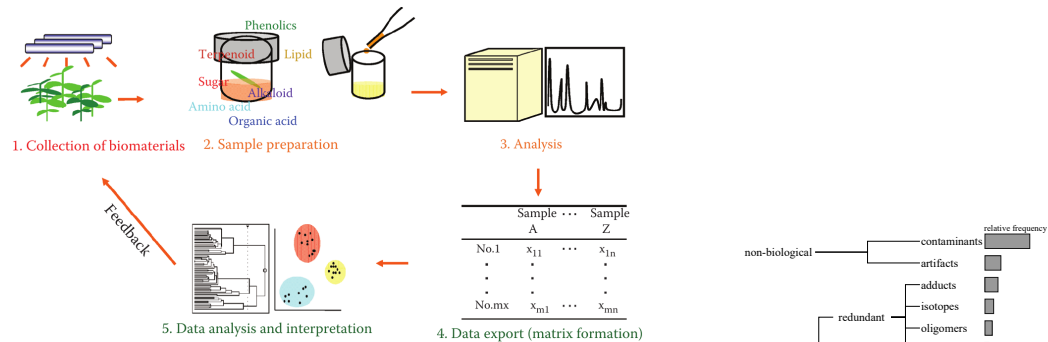


Figure 1: Metabolomics workflow ^a.

^aVeličković, P., Cucurull, G., Casanova, A., Romero, A.,
Edwin A. M. et al.

Figure 2: Untargeted

Our objectives

Aligning metabolomic networks to filter relevant features from Mass Spectrometry data via Graph Neural Networks.

- Develop a pipeline for the analysis of metabolomic networks.
- Model metabolomic networks from mass spectrometry data.
- Implement GNN models to learn metabolite representations.
- Evaluate the effectiveness of the proposed GNN-based alignment approach.

Graph Neural Network (GNN)

It is a function $f: (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ ($d \ll |V|$).

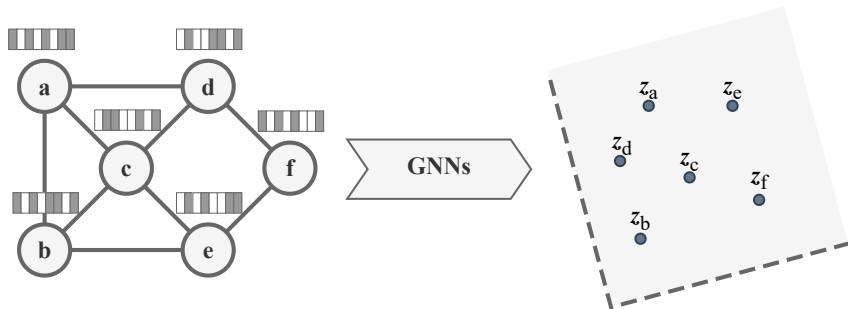


Figure 3: GNNs scheme.

Graph alignment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V}_1 \rightarrow \mathbf{V}_2$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

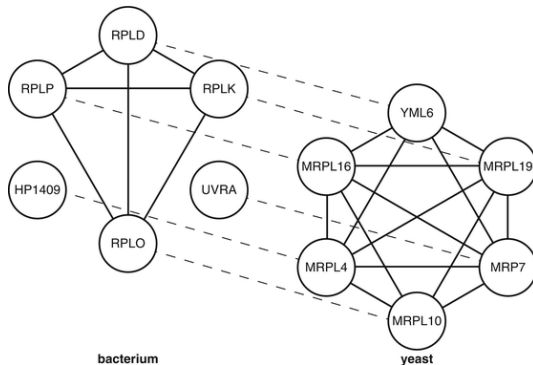


Figure 4: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

Base GNN model

T-GAE: Transferable Graph Autoencoder for Network Alignment [?].

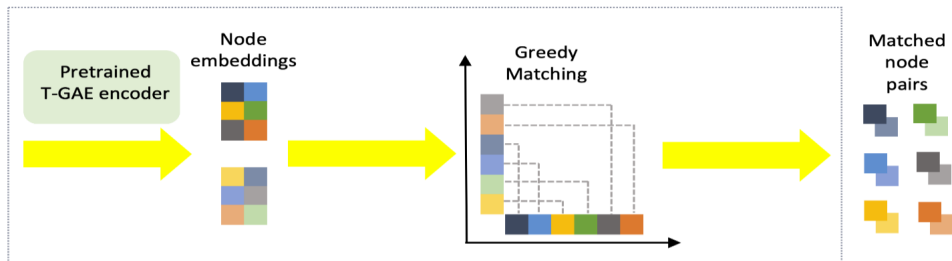


Figure 5: T-GAE encoder.

Mass spectrometry data

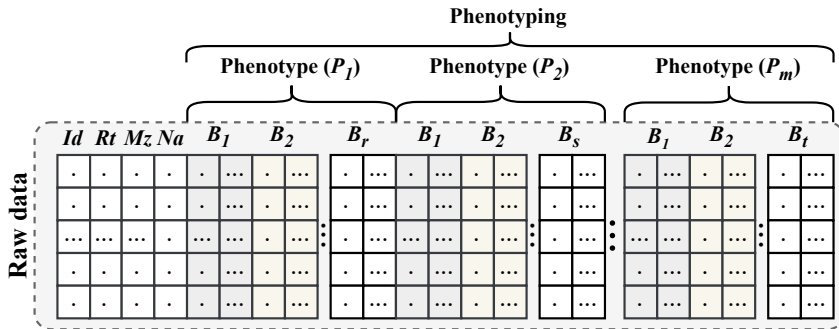


Figure 6: Dataset format.

Method

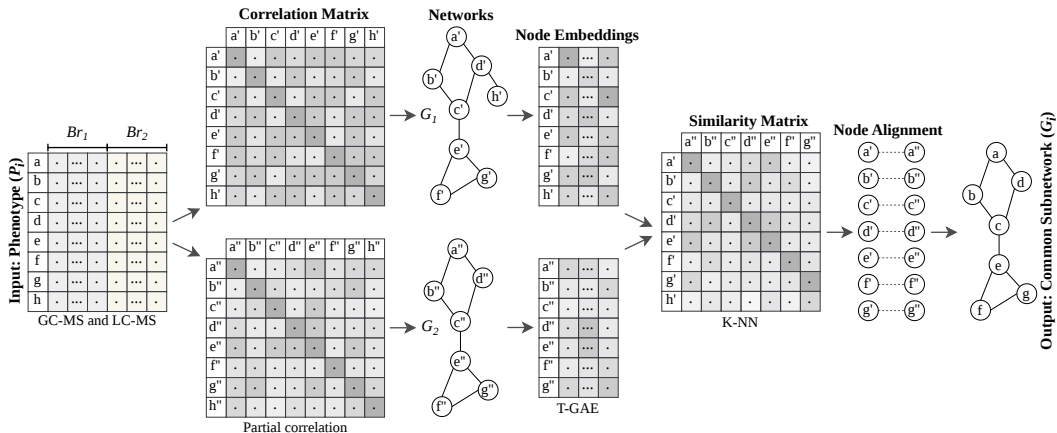


Figure 7: Our workflow.

Experiments setup

The experiments were run on an AMD Ryzen Threadripper PRO 5955WX computer with 128 cores, 256 GB of RAM, 2x NVIDIA RTX A6000 with 48 GB of VRAM. The source code was implemented in Python with Scikit-learn and PyTorch Geometric.

Table 1: Experiments setup.

Features	Values
GNNs architectures	T-GAE with GIN, GINE encoders
Embedding dimensions	16, 32, 64
Optimizer	Adam with $lr = 10^{-4}$, $weight_decay = 5 \times 10^{-4}$
Early stopping	$patience = 10$
Epochs	100
Node features	Mz , Rt , $\log(Avg\ intensities)$, $Std\ intensities$, $Cv\ intensities$
Edge features	$ \rho_{ij} $, $sign(\rho_{ij})$, ρ_{ij}^2

Dataset Cocoa, Mentos, Leaf, Synthetic

Dataset

These 4 datasets were used in the experiments.

- Cocoa (from ICOBA-PUCP)
- Mentos (from ICOBA-PUCP)
- Leaf [?]
- Synthetic

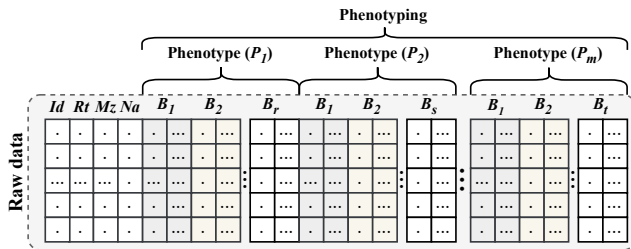


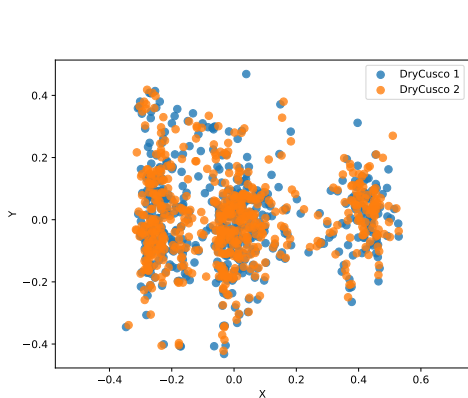
Figure 8: Mass spectrometry data format.

Build metabolomic networks

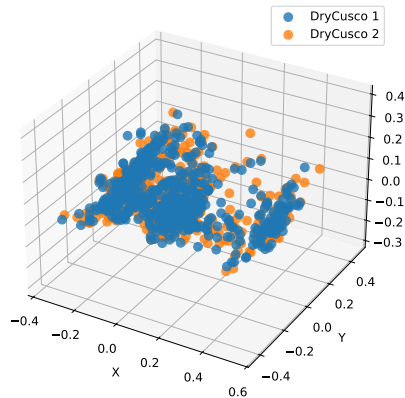
Table 2: Cocoa dataset.

Group	Subgroup	$ V $	$ E $	Density	Is connected?
DryAmazonas	1	559	103 333	0.662557	True
DryAmazonas	2	559	104 793	0.671918	True
DryCusco	1	559	102 044	0.654292	True
DryCusco	2	559	102 392	0.656523	True
DrySanMartin	1	559	102 897	0.659761	True
DrySanMartin	2	559	102 453	0.656914	True
FreshAmazonas	1	559	105 331	0.675368	True
FreshAmazonas	2	559	105 722	0.677875	True
FreshCusco	1	559	103 348	0.662653	True
FreshCusco	2	559	104 638	0.670924	True
FreshSanMartin	1	559	104 932	0.672809	True
FreshSanMartin	2	559	105 894	0.678977	True

Node embeddings (T-GAE)



(a) For Cocoa (DryCusco).



(b) For Cocoa (DryCusco).

Figure 9: Node embeddings.

Network alignment

Table 3: Node alignments (DryCusco).

ID 1	ID 2
0	93
2	2
4	57
5	5
7	93
...	...
542	542
551	551
556	509
557	550
558	530

Table 4: Node filtered **294/559** (DryCusco).

ID	Rt	Mz	Formula
2	0.097	172.95681	Unknown
5	0.201	144.98254	Unknown
15	1.871	118.08675	C5 H11 N O2
38	2.086	381.08003	C14 H16 Cl F3 N4 O S
46	2.110	118.08671	C5 H11 N O2
...	...	...	...
531	29.792	374.30350	C24 H36 O2
537	29.838	454.42642	C28 H55 N O3
538	29.839	150.12816	C10 H15 N
542	29.860	398.23344	C24 H31 N O4
551	29.954	150.12807	C10 H15 N

Comparison of encoders

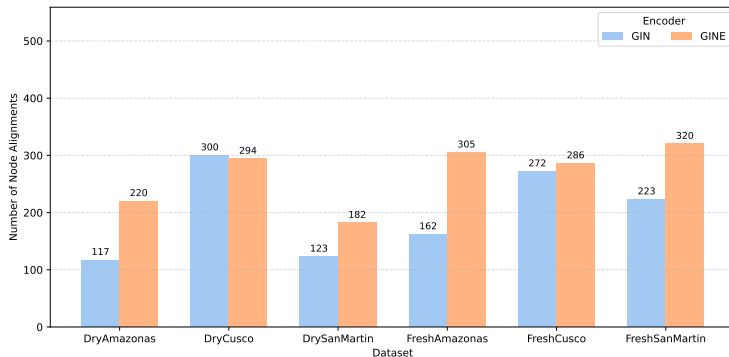


Figure 10: GIN vs. GINE.

Conclusions

Our proposal, is a approach to the exploratory analysis of metabolic networks.

- Features shared by Cocoa beans within the same group are retained, reducing the number of features before identification is attempted.
- A core graph embedding is generated for each Cocoa bean, providing a compact representation that can be used as input to a classifier for subsequent Cocoa bean identification.
- The alignment workflow enables samples acquired on different days to be integrated into a single dataset. This approach mitigates known batch effects, such as retention-time shifts, while the graph embedding favors correlations among signals rather than depending on their absolute intensity values.

Thanks!
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<https://github.com/win7>